

TNFRSF9 (CD137/4-1BB) is an activation-induced gene in human cutaneous mast cells

Absent at rest, switched on ~60-fold by IL-33 and IgE cross-linking, and modestly and heterogeneously elevated in atopic dermatitis skin

Analysis of six public human transcriptomic cohorts. All figures, legends and statistics are reproducible from analysis/ in the project repository.

Subject: cutaneous mast cells. Fibroblasts, keratinocytes, T/NK cells and macrophages appear in this report in exactly one role — as control populations, to show that a mast-cell result is specific and not a batch or global effect (CLAUDE.md §2).

Figure conventions: error bars are ± 1 SEM throughout. Intervals quoted in the text are 95% confidence intervals and are labelled as such.

Abstract

Mast cells sit at the centre of type-2 skin inflammation, and TNFRSF9 (CD137, 4-1BB) is a co-stimulatory receptor of therapeutic interest, but whether cutaneous mast cells express it — and whether that changes in atopic dermatitis (AD) — has not been established. Using six public human cohorts spanning single-cell, spatial, bulk and in-vitro transcriptomics (20 AD and 12 healthy donors in vivo; primary human skin mast cells in vitro), we find that **TNFRSF9 is an activation-induced gene in mast cells rather than a disease marker**. Resting primary human skin mast cells are essentially TNFRSF9-negative (0.14 FPKM); IL-33 induces the gene 29-fold, IgE-receptor cross-linking 7-fold, and the two together 59-fold, while TSLP and IL-25 alone do nothing. An independent paired-donor cohort reproduces this (IL-33 $\log_2$ FC +6.53, $P = 0.001$, $n = 4$ donors). In AD skin the discovery cohort shows a clear increase — 0.56% of healthy mast cells carry a TNFRSF9 transcript versus 2.91% in AD (Fisher OR 5.35, $P = 0.001$) — but **this does not reproduce in the one independent cohort with adequate mast-cell recovery** (2.16% versus 2.45%, OR 1.14, $P = 1.00$), where the healthy baseline is four-fold higher. Mast-cell abundance is unchanged in both, so nothing in this system is driven by cell number. Mast cells nevertheless contribute only 2.7–5.7% of the TNFRSF9 in skin, and whole-tissue TNFRSF9 rises as much in psoriasis as in AD, so the tissue-level signal marks inflamed skin rather than AD. The in-vitro mechanism supplies the likely explanation for the discordant in-vivo cohorts: TNFRSF9 reports how activated a mast cell was when the biopsy was taken, which is not a fixed property of a diagnosis.

1. TNFRSF9 is expressed by cutaneous mast cells, at low level

In the discovery cohort (GSE204762: 11 AD donors with paired non-lesional and lesional biopsies, 6 healthy donors, 280,518 cells, 39 libraries), 3,661 cells passed the mast-cell marker QC of CLAUDE.md §6 (≥ 2 of TPSAB1/TPSB2/CPA3). Of these, 94 (2.57%) carried at least one TNFRSF9 transcript. The replication cohort with adequate mast-cell recovery (GSE153760) gave 33 of 1,379 (2.4%). Expression is a reproducible property of the population, not an artefact of one dataset — but it is a low-expression gene, and every analysis below is governed by counting statistics rather than fold-change estimation (Fig. 1).

Mast cells are also the shallowest-sampled population in the atlas: a median of 546–622 UMI per cell against 3,157–4,061 for fibroblasts and keratinocytes. Section 7.1 sets out what follows from that, and why per-cell UMI is *not* "sequencing depth".

2. Healthy versus AD: the primary comparison (CLAUDE.md §3)

2.1 Mast-cell numbers and TNFRSF9⁺ mast-cell numbers

Each point in Fig. 2 is one sample (one donor $\times$ arm); statistics are donor-level Mann–Whitney against the healthy arm, plus a pooled Fisher exact test on raw TNFRSF9⁺ cell counts.

Quantity	Healthy	AD non-lesional	AD lesional	MWU vs healthy (NL / LS)
Samples (donors)	6 (6)	10 (10)	11 (11)	—
Cells profiled	59,745	97,947	122,826	—
Mast cells recovered	538	1,564	1,559	—
Mast cells per sample, median	99.5	131	151	$P = 0.37 / 0.26$
Mast cells, % of all cells (median)	1.47%	1.30%	1.21%	$P = 0.79 / 1.00$
TNFRSF9⁺ mast cells	3	52	39	—
TNFRSF9 ⁺ per sample, median	0	2.5	2.0	$P = 0.036 / 0.106$
TNFRSF9⁺ as % of mast cells	0.56%	3.33%	2.50%	$P = 0.020 / 0.090$

Pooled healthy versus AD: 3 of 538 versus 91 of 3,123 mast cells are TNFRSF9⁺ (Fisher OR 5.35, $P = 5.3 \times 10^{-4}$).

Read together, as CLAUDE.md §4 requires: **mast-cell abundance does not differ between healthy and AD skin (1.47% versus 1.30%/1.21% of cells, $P = 0.79/1.00$), while the fraction of mast cells carrying TNFRSF9 rises about 5-fold (0.56% to 2.50–3.33%).** The change is in what mast cells express, not in how many there are. Five of six healthy donors contributed no

TNFRSF9⁺ mast cell at all.

A per-transcript version of the same contrast — TNFRSF9 molecules per mast cell, which uses all 110 transcripts rather than only presence/absence — gives log₂FC +2.40 (6.1×, exact donor permutation P = 0.030, n = 6 vs 11 donors); normalising by mast-cell UMI instead gives +2.81 (P = 0.012). Section 7.1 explains why the per-cell version is the one quoted.

2.2 Healthy versus AD non-lesional, and versus AD lesional

Comparison	log ₂ FC (per mast UMI)	Exact permutation P	Transcripts	Donors
Healthy vs AD non-lesional	+3.05	0.0031	3 vs 58	6 vs 10
Healthy vs AD lesional	+2.59	0.179	3 vs 49	6 vs 11

Both arms of AD skin sit above healthy skin. The lesional comparison misses significance despite a similar effect size because lesional mast cells vary more between donors — a power statement, not evidence of absence.

2.3 The refinement: AD lesional versus non-lesional

Within patients (10 donors with both samples), per-mast-cell TNFRSF9 did not differ between lesional and non-lesional skin (log₂FC −0.46, 95% CI −1.97 to 1.06, P = 0.55). Mast-cell TNFRSF9 is a property of AD skin generally rather than of the lesion — consistent with 2.2, where the non-lesional contrast is the stronger of the two.

3. Replication: the AD increase does not reproduce (CLAUDE.md §7)

Which cohorts can test this. A cohort contributes only if it recovered ≥100 marker-QC mast cells in each arm and ≥0.5% of its cells as mast cells — a mast-cell QC threshold judged on tryptase/CPA3 marker data alone, independent of TNFRSF9 and of the outcome.

Cohort	Mast-cell recovery	Mast cells H / AD	Qualifies
GSE153760	4.41%	278 / 1,101	yes
GSE204762 (discovery)	1.31%	538 / 3,123	yes
GSE222840 + GSE173205	0.106%	27 / 91	no

GSE222840+GSE173205 recovered mast cells 12–40× less efficiently than the other two and yielded 2 TNFRSF9 transcripts in 111,370 cells (Fig. 5d). A cohort that cannot recover the cell type cannot test a gene in it, so it is reported as an attempted replication that failed on mast-cell recovery, not as evidence about TNFRSF9, and it is not pooled.

The result in the one qualifying independent cohort.

Cohort	TNFRSF9 ⁺ healthy	TNFRSF9 ⁺ AD	Fisher OR	P
Discovery GSE204762	3/538 (0.56%)	91/3,123 (2.91%)	5.35	0.001
GSE153760	6/278 (2.16%)	27/1,101 (2.45%)	1.14	1.00

The AD increase does not reproduce. In GSE153760 the TNFRSF9⁺ mast-cell fraction is the same in healthy and AD skin. The discrepancy is not in the AD arms, which agree closely (2.91% versus 2.45%); it is in the healthy arms, which differ four-fold (0.56% versus 2.16%). GSE153760's healthy baseline leaves no room for an increase.

Neither cohort is well placed to settle which healthy baseline is right: GSE153760 has **two** healthy donors, one contributing 5 of its 6 TNFRSF9⁺ cells; the discovery cohort has six, five of them contributing none.

On the per-transcript rate the two cohorts agree in direction (discovery log₂FC +2.40, GSE153760 +0.58), and pooled they give +1.27 (95% CI 0.33 to 2.22, P = 0.0084) under fixed effects and +1.43 (95% CI −0.36 to 3.21, P = 0.12) under random effects. But the primary quantification — what fraction of mast cells carry the transcript — replicates only as *expression*, not as an *AD-associated increase*.

What replicates: that cutaneous mast cells express TNFRSF9, at 2.2–2.9% of mast cells in both qualifying cohorts.

What does not: the increase in AD.

4. Mechanism: TNFRSF9 is induced by the two signals that define AD

No tissue analysis can separate "mast cells transcribe more TNFRSF9 in AD" from "AD skin holds a different mast-cell population". A stimulation experiment can, and two public datasets perform exactly that on **primary** human mast cells — with no dissociation artefact, no ambient RNA, no doublets and no depth confound (Fig. 6).

4.1 Primary human SKIN mast cells + the AD milieu (GSE196862)

Skin-derived mast cells, 24 libraries, stimulated with IgE-receptor cross-linking and the epithelial alarmins IL-33, TSLP and IL-25, alone and combined.

Condition	TNFRSF9 (FPKM)	log ₂ FC vs resting
resting	0.14	—
TSLP	0.11	-0.28
IL-25	0.11	-0.32
IgE/Ag	1.07	+2.87 (7×)
IL-33	4.33	+4.88 (29×)
IL-33 + IL-25 + TSLP	3.76	+4.67
IgE/Ag + IL-33	8.66	+5.88 (59×)
IgE/Ag + all three alarmins	7.96	+5.75

Resting skin mast cells are effectively TNFRSF9-negative (replicates 0.011, 0.047, 0.006, 0.487 FPKM). IL-33 and FcεRI cross-linking each induce the gene and synergise; TSLP and IL-25 alone do nothing, so this is not a generic cytokine response.

Specific, not global. In the same libraries housekeeping genes are flat (B2M +0.34, RPL13A -0.23, TMSB4X +0.19 under IgE+IL-33) and mast identity genes barely move (TPSAB1 -0.29, TPSB2 +0.02). Among co-stimulatory TNF-receptor relatives, TNFRSF9 has both the lowest resting level (0.14 FPKM versus 1.6–1.8 for TNFRSF18 and TNFRSF4) and the largest induction (+5.88 versus +3.20 and +0.73): the most switch-like member of the family in this cell type.

4.2 Primary human mast cells + activated CD4⁺ T cells (GSE235240)

Four donors, paired, mast cells FACS-sorted after 24 h co-culture.

Condition	TNFRSF9 (normalised counts)	paired log ₂ FC	P	n donors
resting	103	—	—	4
resting CD4 ⁺ T cells	133	+0.26	0.17	4
activated CD4⁺ T cells	768	+2.97	0.003	4
IgE/Ag	4,148	+5.51	0.001	4
IL-33	9,079	+6.53 (92×)	0.001	4

The ordering reproduces with an independent stimulus and a paired design: IL-33 > IgE/Ag > activated T cells > resting T cells ≈ unstimulated. Tryptase is unchanged (TPSAB1 log₂FC -0.04, P = 0.61, n = 4 donors), so this is not a differentiation or viability effect.

4.3 What this settles

The mechanism. TNFRSF9 in mast cells is switched on from a near-zero baseline by exactly the two signals that define AD: IL-33 released by damaged keratinocytes, and IgE cross-linking on FcεRI. This is a *measured* result in primary cells, not an inference from tissue.

Why the in-vivo data look as they do. If TNFRSF9 reports activation state rather than disease identity, then healthy mast cells should be near-zero (they are); AD mast cells should be positive to a degree depending on how activated they were at biopsy, predicting the observed between-cohort heterogeneity (Section 3); and the tissue-level rise should not be AD-specific, because any mast-activating inflammation would do the same (Section 6).

What it does not settle. These are in-vitro stimulations, not AD skin. They establish that mast cells can express TNFRSF9 on demand and identify sufficient stimuli. They do not establish how much of the in-vivo signal runs through this pathway, nor that 4-1BB protein reaches the cell surface, nor any function.

5. In situ, without dissociation (GSE197023 Visium)

The per-cell estimate is conditional on which mast cells survive enzymatic digestion (CLAUDE.md §4). Visium never dissociates the tissue and reproduces the direction: whole-spot TNFRSF9 was higher in AD than healthy skin (log₂FC +2.06, 95% CI 0.35–3.77, P = 0.019; 13,742 spots, 6 healthy versus 13 AD sections, donor-clustered with a log-depth offset), driven by lesional skin (+2.30, 95% CI 0.47–4.12, P = 0.014). Mast-cell content in situ was unchanged or slightly lower (TPSAB1 log₂FC -0.80, P = 0.34; CPA3 -0.71, P = 0.021, 6 vs 13 sections), matching the unchanged abundance in the single-cell arm.

Spot depth differs ~4-fold between arms (AD lesional 4,025 versus healthy 1,220 UMI), so every section in Fig. 3 is annotated with its own median depth.

The §6 control genes were carried through the identical group test: COL1A1 (fibroblast) and KRT14 (keratinocyte) showed no comparable rise, so the spatial result is not a global shift (results/tables/spatial_group_glm.csv).

6. Mast cells are a minor source of skin TNFRSF9, and the tissue signal is not AD-specific

Arm	TNFRSF9 in mast cells	in whole tissue	mast share
Healthy	3	104	2.9%

Arm	TNFRSF9 in mast cells	in whole tissue	mast share
AD non-lesional	58	1,018	5.7%
AD lesional	49	1,844	2.7%

Mast cells account for 0.22–0.41% of the tissue transcriptome and 2.7–5.7% of its TNFRSF9. Induction within mast cells and their small share of the total are both true; a receptor switched on in a rare cell can matter functionally without dominating a bulk measurement, but neither can it explain a tissue-level signal.

Whole-skin RNA-seq (GSE121212, 147 samples) makes the specificity problem explicit:

Comparison	log ₂ FC	95% CI	n	P
Healthy vs AD	+2.48	1.85 to 3.11	38 vs 54	1.0×10 ⁻¹⁴
Healthy vs AD non-lesional	+1.38	0.75 to 2.01	38 vs 27	1.6×10 ⁻⁵
Healthy vs AD lesional	+3.06	2.37 to 3.75	38 vs 27	2.9×10 ⁻¹⁸
AD lesional vs non-lesional	+1.68	1.04 to 2.31	27 vs 27	2.4×10 ⁻⁷
Healthy vs psoriasis	+2.20	1.80 to 2.59	38 vs 55	8.7×10 ⁻²⁸

Poisson GLM on raw counts, log-library-size offset, errors clustered on patient. Mast-cell content in the same samples was flat (TPSAB1 log₂FC -0.09, P = 0.74; TPSB2 -0.10, P = 0.73; CPA3 -0.48, P = 0.037, n = 38 vs 54). Nor does whole-skin TNFRSF9 track mast-cell content *within* any arm (Spearman ρ = +0.20 healthy, +0.26 AD non-lesional, +0.12 AD lesional, -0.13 psoriasis lesional; all P > 0.18, n = 27–38 samples per arm) — consistent with mast cells being a minor source (Section 6 above). In bulk tissue, abundance and per-cell expression are not separable at all, and the mast-cell contribution is not separable from any other source. What bulk establishes is that skin TNFRSF9 rises steeply in AD, does so without any rise in mast-cell content, and rises just as much in psoriasis — a marker of inflamed skin.

7. Rigour: estimand, controls, artefacts, calibration

7.1 What "depth" means, and which estimand follows

Sequencing depth is a property of the run and is shared by every cell in a library. Per-cell UMI is not that: it is library reads × capture efficiency ×

the cell's own mRNA content. Only the first term is sequencing, and normalising by per-cell UMI divides out the third, which is biology.

Variance decomposition of log(UMI per cell) across mast cells and the control populations:

Component	Share	Can it be sequencing depth?
Between libraries (flow cell, loading)	27.4%	Yes
Between cell types within a library	22.0%	No — those cells shared a flow cell
Cell to cell within a cell type	50.6%	No

Within-library ratios make this concrete: mast cells yield 0.08–0.30 of the UMI that fibroblasts and keratinocytes yield **in the same run**, in all 39 libraries, and that ratio is 34% lower in AD (median 0.134, n = 32 libraries) than healthy (0.203, n = 7 libraries; Mann–Whitney P < 0.001). Decisively, the two move in opposite directions: **AD libraries are sequenced more deeply overall (median 4,292 versus 1,564 UMI per cell) yet AD mast cells are shallower (693 versus 935).** No flow-cell effect can do that. AD mast cells are transcriptionally smaller — 399 versus 477 genes detected, less tryptase per cell.

Consequence. Dividing by mast-cell UMI removes biology along with technical effort, using a denominator 25% smaller in AD. The **per-cell estimand is primary** here; the per-UMI estimand is reported beside it as the compositional companion, as CLAUDE.md §6 requires. The result does not depend on the choice:

Estimand	Healthy	AD	Fold	P
per mast UMI (adjusted)	0.057 /10k	0.469 /10k	8.2×	0.012
per mast CELL (unadjusted)	0.56 /100	3.43 /100	6.1×	0.030
no normalisation (TNFRSF9⁺ cell counts)	3/538	91/3,123	OR 5.35	5.3×10⁻⁴

But the adjustment cannot be dropped for the control populations, which are ~2× deeper in AD. Unadjusted, TNFRSF9 appears to rise in almost every cell type — the "shared shift means batch" pattern CLAUDE.md §6 exists to catch:

Population (6 healthy vs 11 AD donors)	log ₂ FC per UMI (adjusted)	log ₂ FC per cell (unadjusted)
Mast (subject)	+2.81 (P = 0.012)	+2.40 (P = 0.030)
Fibroblasts	+0.91 (P = 0.23)	+2.05 (P = 0.011)
Keratinocytes	+0.26 (P = 0.72)	+1.37 (P = 0.15)
T/NK	+1.49 (P = 0.0001)	+2.06 (P = 0.0013)

Population (6 healthy vs 11 AD donors)	log ₂ FC per UMI (adjusted)	log ₂ FC per cell (unadjusted)
Macrophages	+2.06 (P = 0.024)	+3.09 (P = 0.013)

The adjustment is unnecessary for the mast-cell effect and essential for the claim that the effect is specific. Both are reported.

7.2 The mandated controls (CLAUDE.md §6)

Control	Result	Consequence
Library size	log-depth offset throughout; depth-stratified analysis below	Effect survives and strengthens
Donor aggregation	All inference donor-level; exact permutation over all 12,376 donor-label assignments	Cluster-robust SEs on 6 clusters not relied upon
Same test in fibroblasts + keratinocytes	+0.91 (P = 0.23) and +0.26 (P = 0.72), 6 vs 11 donors	No shared shift → not batch
Per-group median depth	Mast 546 / 541 / 622 UMI; tissue 1,999 / 2,833 / 3,306 (healthy / AD NL / AD LS)	Mast depth comparable across arms
Marker QC	3,661 of 4,313 deposited "Mast" labels passed (84.9%); 433 QC-positive cells carried non-mast labels and were excluded	Deposited labels imperfect, as anticipated
Dissociation stress	HSP+IEG 986 per 10k mast UMI healthy versus 416 AD	Healthy is the most stressed arm — biases against the finding ; Spearman(stress, rate) = -0.15, P = 0.57

Sampling-matched detection. Stratifying mast cells by UMI per cell, healthy mast cells contributed zero TNFRSF9 transcripts in every stratum below 800 UMI, where AD mast cells contributed 35. The stratified common rate ratio is **9.6×**, larger than the crude estimate (results/tables/sc_depth_matched.csv).

Leave-one-donor-out. Dropping any one of the 17 discovery donors (6 healthy, 11 AD) leaves log₂FC between +2.15 and +4.72, permutation P ≤ 0.043 throughout. Removing the most influential donor (MGH108) gives +2.15 (P = 0.023).

7.3 The doublet artefact

T/NK cells carry TNFRSF9 at a rate comparable to AD mast cells, so a mast–T doublet would present as exactly what is being counted. TNFRSF9⁺ mast cells do carry T-cell transcripts more often than TNFRSF9[−] ones (10.6% versus 3.4%; depth-stratified OR 2.68 across six strata, n = 94 versus 3,567 cells), so some contamination exists. No cell was flagged by the deposited scrublet call, and TNFRSF9⁺ mast cells had *lower* doublet scores (median 0.02 versus 0.03).

The decisive test is deletion:

Cell set	Mast cells	TNFRSF9 H / AD	log ₂ FC	Permutation P
All marker-QC mast cells	3,661	3 / 107	+2.81	0.012
Excluding any cell with a T-cell transcript	3,531	3 / 97	+2.73	0.022
Excluding any T-cell or myeloid transcript	3,187	3 / 78	+2.40	0.050

97 of 107 AD transcripts come from mast cells with **no T-cell transcript at all**, so doublets cannot be the source. The stricter purge removes 13% of mast cells and pushes the test to the boundary: the effect is not eliminated, but this cohort has no power to spare.

7.4 Calibration: is the effect exceptional?

- **Is the test calibrated?** Permuting arm labels *within* AD donors, where no disease contrast exists by construction, gives a median 4.0% of matched genes at P ≤ 0.05 (IQR 2.9–5.6%, 40 random splits). The exact permutation test is approximately calibrated.
- **Is TNFRSF9 exceptional?** No. The real contrast yields 17.8% of matched genes at P ≤ 0.05 — far outside the null range — so the mast-cell transcriptome differs widely between healthy and AD skin. Against that background TNFRSF9 ranks 290 of 3,454 (empirical P = 0.084, Fig. S1b).

TNFRSF9 induction in AD mast cells is a real, calibrated nominal result in the top 8% of a broad disease-associated shift — but it is *one of many* genes moving, and the claim that TNFRSF9 is *specifically* dysregulated relative to the rest of the mast-cell transcriptome is **not** supported.

8. Measured, modelled, inferred (CLAUDE.md §8)

- **Measured.** Transcript counts per cell, per spot, per bulk library and per in-vitro condition; marker expression; sequencing depth; the in-vitro induction of TNFRSF9 by IL-33 and IgE in primary skin mast cells.
- **Modelled.** All fold changes and intervals (Poisson/binomial GLMs with log-depth offsets and donor- or patient-clustered errors; exact donor-label permutation; inverse-variance meta-analysis).

- **Inferred.** That the in-vivo per-mast-cell difference reflects the same activation programme demonstrated in vitro; and that it is transcriptional induction rather than selective survival of a TNFRSF9⁺ subset through dissociation. The Visium arm supports the second indirectly but cannot resolve single mast cells at 55 µm.
- **Not established.** 4-1BB protein on the mast-cell surface (see `results/PROTEIN_EVIDENCE.md`); any functional consequence; AD-specificity of the tissue-level change; the in-vivo magnitude ($I^2 = 42\%$); and selective dysregulation relative to the rest of the mast-cell transcriptome.

9. Limitations

1. **Small counts.** Nine TNFRSF9 transcripts across all healthy mast cells in all three in-vivo cohorts. Statements about healthy skin are upper bounds.
2. **Donor numbers.** 12 healthy and 20 AD donors in vivo, with each cohort's signal concentrated in one or two donors.
3. **Mast-cell recovery governs everything.** Recovery ranged 0.11%–4.4% of cells across cohorts under one rule. One dedicated AD atlas (GSE147424, cryopreserved biopsies) contained *no* recoverable mast cells and was excluded (`docs/datasets.md`); suction-blister sampling also fails to capture them.
4. **The AD increase is not replicated.** It is clear in the discovery cohort (OR 5.35, $P = 0.001$) and absent in the one qualifying independent cohort (OR 1.14, $P = 1.00$). The disagreement is in the healthy baseline (0.56% vs 2.16% of mast cells), which neither cohort has the donors to settle.
5. **Not exceptional against background** (7.4), and **not AD-specific** at tissue level (Section 6).
6. **In vitro is sufficiency, not contribution.** Section 4 shows mast cells *can* be driven to express TNFRSF9; it does not quantify how much of the AD signal runs through that route.
7. **Transcript, not protein.** Everything above is RNA. Protein-level evidence is assessed separately in `results/PROTEIN_EVIDENCE.md`.

Figures

Conventions used throughout. Error bars show the standard error of the mean. Intervals given in the report text are 95% confidence intervals and are labelled as such. Mast cells were identified by marker quality control, requiring at least two of TPSAB1, TPSB2 and CPA3 to be detected in a cell. All group comparisons were made at the level of donors or samples, never individual cells. Statistical tests are two-sided.

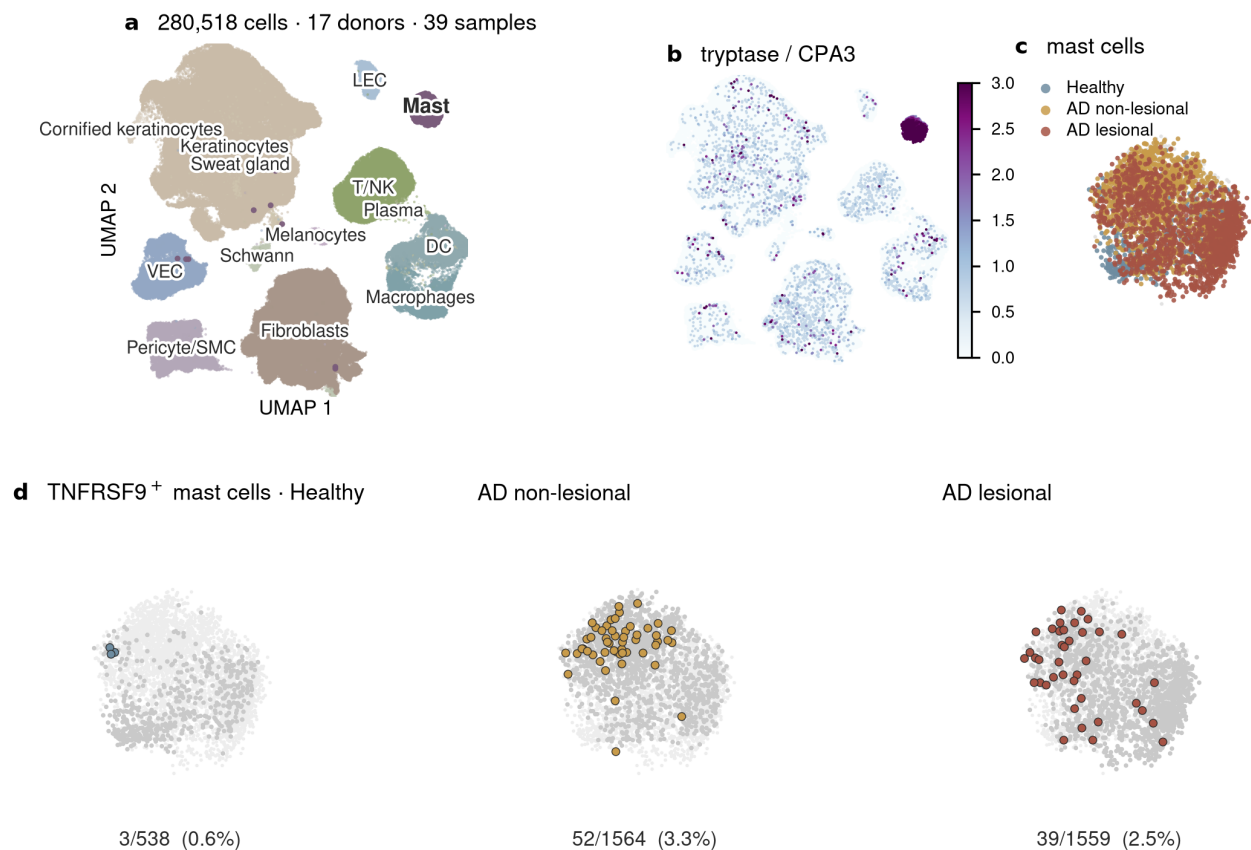


Figure 1 | Cutaneous mast cells and TNFRSF9 in the discovery cohort

a, Uniform manifold approximation and projection (UMAP) of 280,518 cells from 39 whole-skin biopsies taken from 17 adults (6 healthy donors and 11 patients with atopic dermatitis, each sampled at lesional and non-lesional sites; GSE204762). Cells are coloured by the cell-type annotation supplied with the dataset. Mast cells are shown in purple.

b, The same embedding coloured by the combined counts of the mast-cell granule genes TPSAB1, TPSB2 and CPA3. The mast-cell island is the only region of the atlas in which these transcripts are abundant.

c, The 3,661 cells that passed mast-cell marker quality control, coloured by tissue group.

d, Mast cells from each tissue group, with TNFRSF9-positive cells shown as filled circles. Values beneath each panel give the number of TNFRSF9-positive mast cells, the total number of mast cells and the corresponding percentage. Three of 538 mast cells were positive in healthy skin (0.6%), compared with 52 of 1,564 in non-lesional atopic dermatitis (3.3%) and 39 of 1,559 in lesional skin (2.5%).

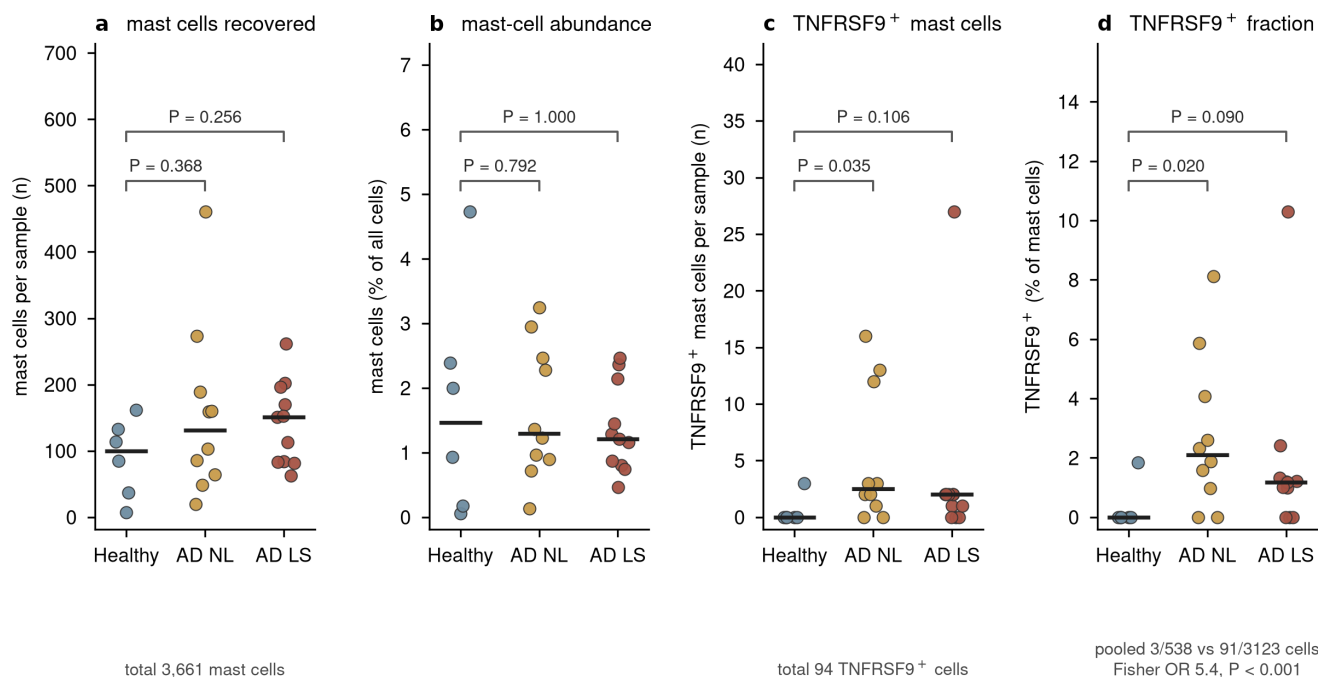


Figure 2 | Mast-cell numbers and TNFRSF9-positive mast-cell numbers

Quantification of the cells shown in Fig. 1d. Each point is one sample, defined as one donor at one tissue site (6 healthy, 10 non-lesional and 11 lesional samples). Horizontal bars show group medians. P values above the brackets are from Mann-Whitney U tests against the healthy group.

a, Mast cells recovered per sample. Recovery was similar across groups (medians of 99.5 cells in healthy skin, 131 in non-lesional and 151 in lesional atopic dermatitis; $n = 6, 10$ and 11 samples; $P = 0.37$ and $P = 0.26$).

b, Mast cells as a percentage of all cells recovered from the same sample. Abundance did not differ between groups (median 1.47% in healthy skin, 1.30% in non-lesional and 1.21% in lesional atopic dermatitis; $P = 0.79$ and $P = 1.00$).

c, TNFRSF9-positive mast cells per sample.

d, TNFRSF9-positive mast cells as a percentage of the mast cells in the same sample. Pooling cells across samples, 3 of 538 mast cells were positive in healthy skin and 91 of 3,123 in atopic dermatitis (odds ratio 5.35, Fisher exact $P = 5.3 \times 10^{-4}$). Five of the six healthy donors contributed no TNFRSF9-positive mast cell.

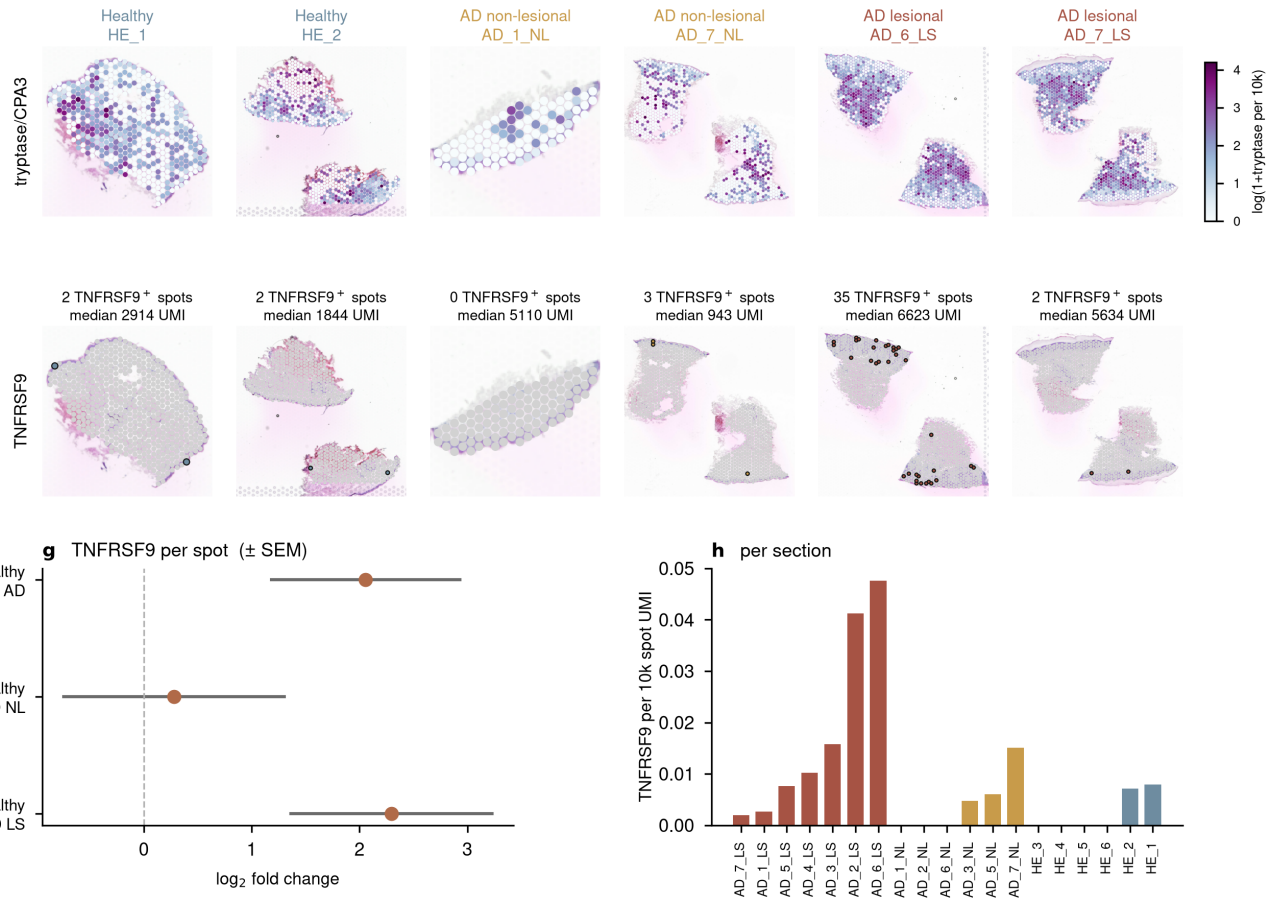


Figure 3 | TNFRSF9 in intact skin measured by spatial transcriptomics

Visium spatial transcriptomics of 19 skin sections from 6 healthy donors and 7 patients with atopic dermatitis (GSE197023), comprising 13,742 spots that passed a minimum depth filter of 200 unique molecular identifiers (UMIs). Sections were not dissociated, so the measurement does not depend on enzymatic release of mast cells. Marker size is scaled to the true spot pitch of each section, which differs between samples because the underlying scans differ in resolution.

Top row, Six representative sections, two per group, with spots coloured by the combined TPSAB1, TPSB2 and CPA3 content per 10,000 UMIs and overlaid on the haematoxylin and eosin image.

Second row, The same sections with TNFRSF9-positive spots highlighted. The number of positive spots and the median sequencing depth are given above each section. Depth per spot differs about fourfold between groups, so it is stated for every section and adjusted for in all models.

g, TNFRSF9 per spot in each comparison, expressed as a log₂ fold change against healthy skin. Estimates come from a Poisson generalised linear model with log UMIs per spot as an offset and standard errors clustered on donor. The fibroblast and keratinocyte genes required as controls were carried through the same test and showed no comparable rise.

h, TNFRSF9 per 10,000 spot UMIs for every section in the cohort.

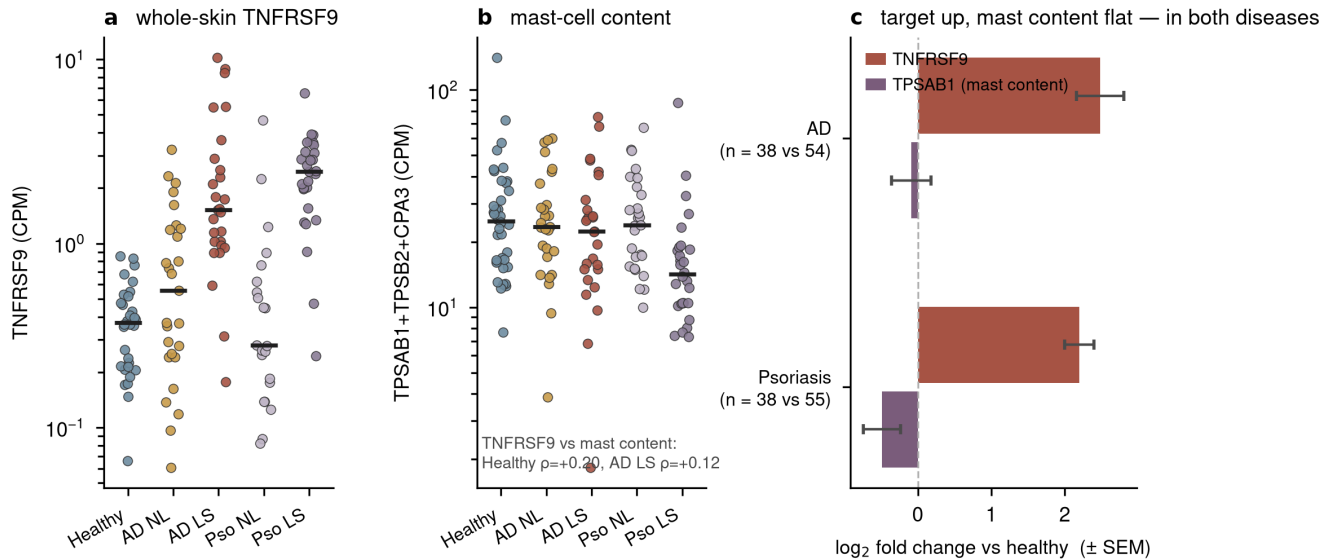


Figure 4 | Whole-skin TNFRSF9 in a large tissue cohort with a disease control

Bulk RNA sequencing of 147 whole-skin biopsies (GSE121212): 38 healthy, 27 non-lesional and 27 lesional atopic dermatitis, and 27 non-lesional and 28 lesional psoriasis. Psoriasis is included as a disease control, to ask whether any change is specific to atopic dermatitis. Each point is one biopsy and horizontal bars show medians.

a, Whole-skin TNFRSF9 in counts per million. Median values rise from 0.36 in healthy skin to 0.56 in non-lesional and 1.52 in lesional atopic dermatitis, and reach 2.46 in lesional psoriasis.

b, Combined TPSAB1, TPSB2 and CPA3 content in the same samples, used here as the tissue-level proxy for mast-cell abundance. Content did not rise with TNFRSF9. Within groups, whole-skin TNFRSF9 showed no correlation with mast-cell content (Spearman rho between -0.13 and +0.26, all $P > 0.18$).

c, Log₂ fold changes against healthy skin for TNFRSF9 and for TPSAB1, shown for atopic dermatitis and psoriasis. Estimates come from a Poisson generalised linear model on raw counts with log library size as an offset and standard errors clustered on patient. TNFRSF9 rose to a similar degree in both diseases while mast-cell content did not change, indicating that the tissue-level signal marks inflamed skin rather than atopic dermatitis specifically.

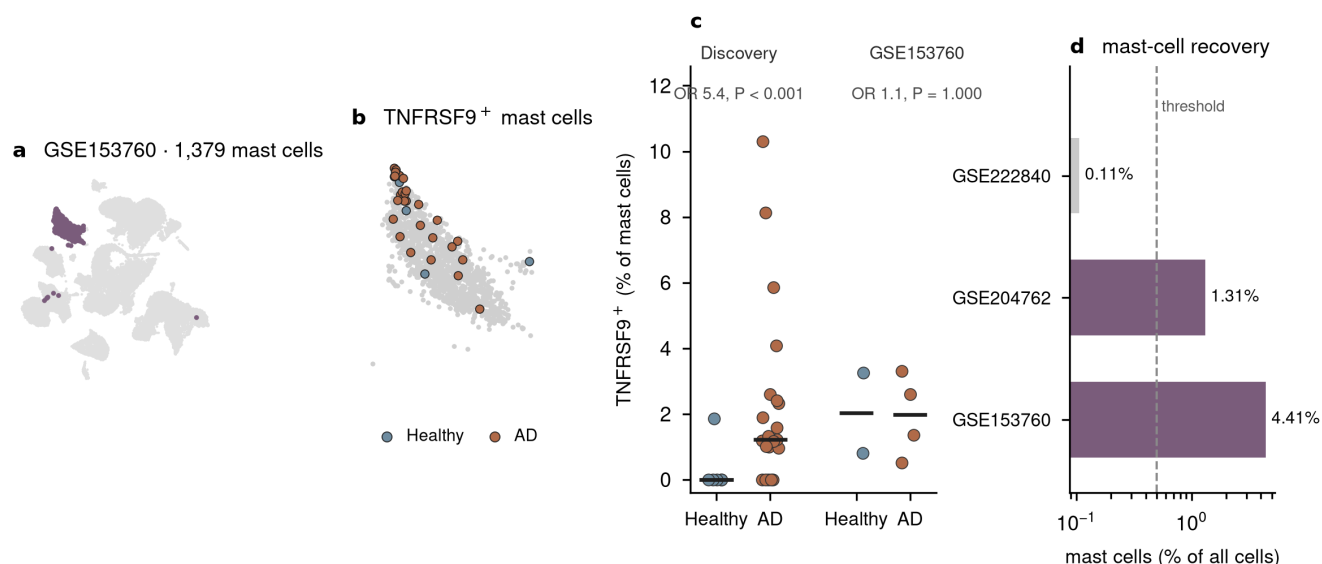


Figure 5 | Replication in an independent cohort

a, UMAP of 31,266 cells from six whole-skin biopsies in GSE153760 (4 patients with atopic dermatitis, 2 healthy donors). The 1,379 cells meeting mast-cell criteria are shown in purple. Suction-blister samples from the same study were excluded because they do not recover mast cells.

b, Mast cells from this cohort with TNFRSF9-positive cells highlighted and coloured by group.

c, TNFRSF9-positive mast cells as a percentage of mast cells, shown for the discovery cohort and for GSE153760 on the same axis. Each point is one sample and horizontal bars show medians. Odds ratios and Fisher exact P values were computed on pooled cell counts within each cohort. The increase seen in the discovery cohort (0.56% to 2.91%, odds ratio 5.35, $P = 0.001$) was not reproduced in GSE153760 (2.16% to 2.45%, odds ratio 1.14, $P = 1.00$). The two cohorts agree on the atopic dermatitis arms and differ fourfold on the healthy arms.

d, Mast-cell recovery for the three single-cell cohorts screened, expressed as the percentage of all recovered cells. The dashed line marks the inclusion threshold of 0.5%. GSE222840 combined with GSE173205 recovered 0.106% of cells as mast cells, giving 27 mast cells in its healthy arm and 2 TNFRSF9 transcripts across 111,370 cells, and was therefore excluded before any TNFRSF9 comparison was made.

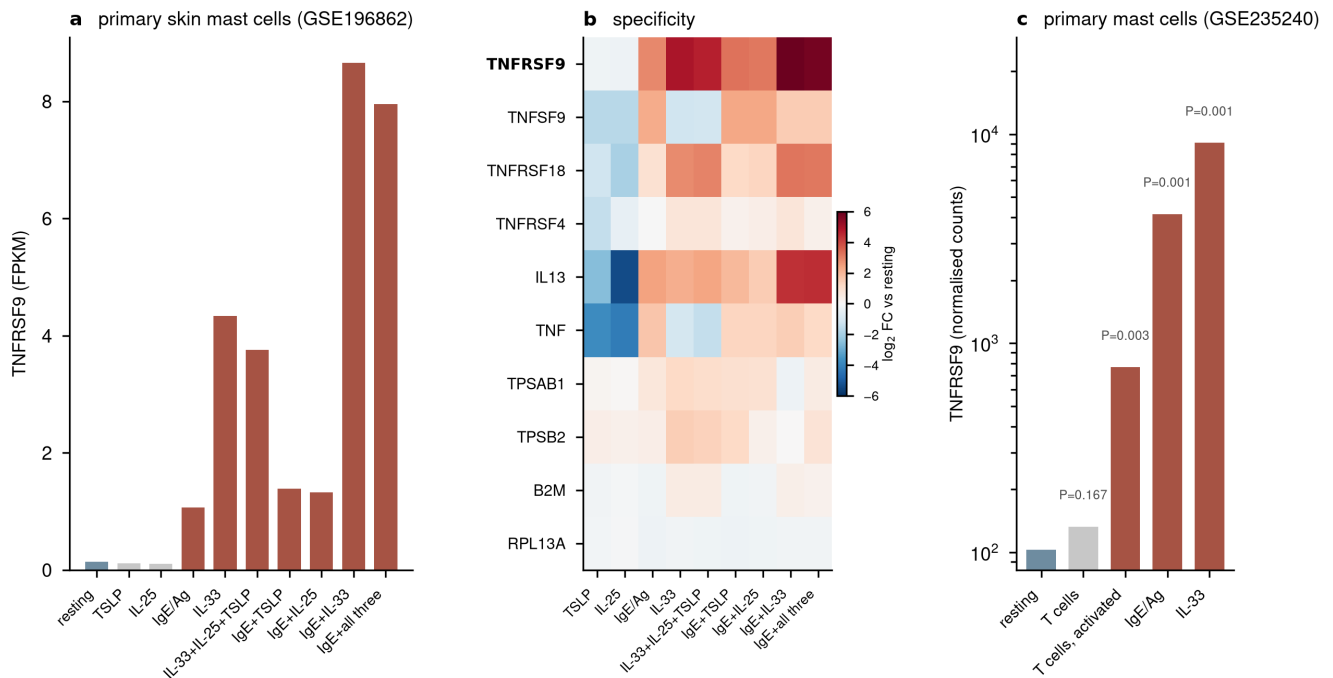
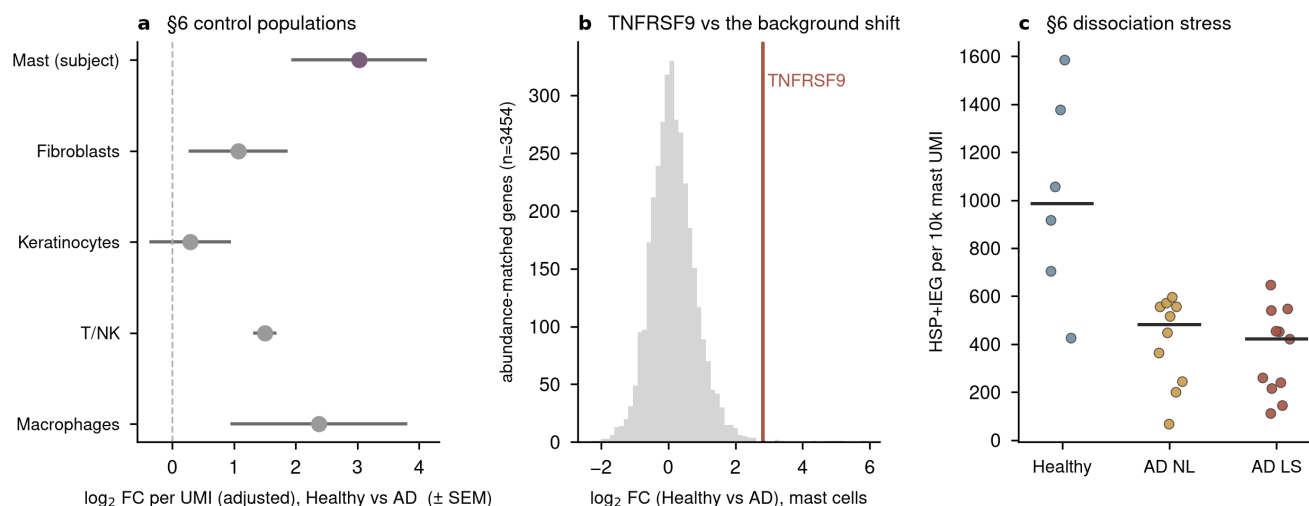


Figure 6 | TNFRSF9 is induced in primary human mast cells by IL-33 and IgE receptor cross-linking

a, TNFRSF9 expression in primary human skin-derived mast cells stimulated for 24 h (GSE196862, 24 libraries, 2 to 4 replicates per condition). Resting cells express almost no TNFRSF9 (0.14 FPKM). IL-33 raised expression 29-fold and IgE receptor cross-linking 7-fold; together they raised it 59-fold, to 8.66 FPKM. TSLP and IL-25 given alone had no effect.

b, Log2 fold changes against resting cells for TNFRSF9, related TNF receptor family members, activation genes, mast-cell identity genes and housekeeping genes. Housekeeping and granule genes were unchanged, so the induction is specific rather than a global shift in the transcriptome.

c, TNFRSF9 in primary human mast cells co-cultured for 24 h with autologous resting or activated CD4⁺ T cells, or stimulated with IL-33 or IgE and antigen (GSE235240). Mast cells were sorted after culture. Four donors were tested and each condition is paired within donor. P values are from paired t-tests on log2 normalised counts. Note the logarithmic axis.



Supplementary Figure 1 | Controls and calibration for the discovery cohort

a, TNFRSF9 in atopic dermatitis compared with healthy skin, tested separately in mast cells and in the control populations required by the project protocol. Estimates are log₂ fold changes from Poisson generalised linear models using each population's own total UMIs as the offset, with standard errors clustered on donor. Fibroblasts and keratinocytes showed no shift, indicating that the mast-cell result is not a batch or global effect.

b, Distribution of log₂ fold changes between healthy skin and atopic dermatitis for 3,454 genes matched to TNFRSF9 for abundance in mast cells. TNFRSF9 is marked. It ranks 290th of 3,454, an empirical P value of 0.084, so it sits in the upper tail of a broad disease-associated shift rather than standing apart from it.

c, Dissociation-stress burden, measured as combined heat-shock and immediate-early gene counts per 10,000 mast-cell UMIs, for each sample. The healthy group carried the highest stress burden. Since stressed cells return lower counts for other transcripts, this works against rather than towards the reported increase in atopic dermatitis.